

And the only storage unit you'll ever need is a USB pen drive.

You can add as many hard drives to this baby as you want. I'd recommend buying some multi-terabyte SATA2 hard drives and attaching them internally. To expand storage, you can use as many external USB HDDs as you want (only if you use USB hubs to increase the number of available USB ports).

You'll need a 256 MB USB pen drive to install the firmware onto this storage unit. Yes, it cannot be called an operating system. What we're talking about is FreeNAS. Head to www.freenas.org and download the latest 60 MB ISO file. Boot up your storage unit with it, and then attach the USB pen drive after FreeNAS has started booting.

Once done, follow these steps to install it:

1. At the first prompt, type "9" (without the quotes) and press **Enter**.
2. At the first screen, select the third choice.
3. Hit **OK** at the next screen.
4. Now select the CD drive (most probably "acd0") where the FreeNAS disk is located.
5. Select the destination drive: your 256 MB or above USB pen drive
6. Do not read any of the text that comes up next (unless it's an error) and hit **Enter**.
7. You're done! Exit the installer now, and reboot the unit. Enter the BIOS and enable booting from the USB (this depends on your BIOS make and version). Remove the FreeNAS disk, keep the USB pen drive attached and reboot the unit. Pretty soon, you'll be booted up into FreeNAS. Congratulations!

By default, you have a static LAN IP address, 192.168.1.250. Leave it at that. Now, open up a Web browser and browse to <http://192.168.1.250>. Now do you know why I called FreeNAS a 'Firmware'?

The default login credentials are "admin" as the username and "freenas" as the password. Once you are in, savour the beauty of the interface for a bit before moving on to configure this as a NFS server.

1. Go to **Disks**→**Management** and click on the "+" icon at the bottom of the empty table.
2. On the resulting page, choose a hard disk (example: ad6). Add a description if you wish. Then, activate SMART monitoring (check the tick box), leave the **Preformatted FS** option as unformatted and hit **Add**.
3. On this page, hit **Apply Changes**. The status column should show **ONLINE**.
4. Now go to **Disks**→**Format**.
5. On this page, select a hard disk (it should have been added in the Management Section). It is important that you use the UFS+SU (GPT) filesystem, as this gives the best speed and reliability. FreeNAS doesn't use an MBR-based partition; it uses the more recent GPT style partition table from the EFI standard. Add a volume label and leave everything else intact. Hit **Format Disk** and then **OK**.
6. The next bit is mounting the partition. Go to **Disks**→**Mount Point**, and hit that "+" icon again. Select a formatted disk, set the **Partition Type** to GPT, the partition number to



Figure 1: FreeNAS system info page

1 and the filesystem to UFS. Add a mount point name, and then remember it. On the next page, hit **Apply Changes**. Disk configuration is now complete. We need to initialise the NFS service.

1. Go to **Services**→**NFS** and check the "Enable" tick box. Set the **Number Of Servers** to something suitable, for example, 16. Then hit **Save And Restart**.
2. Now go to the **Shares** tab and click on the "+" icon.
3. On the resulting page, set a **Path To Share**. This refers to one of your mounted disks. Use the format **/mnt/<mount point name>**. Then select whether to map all users as root (it's safe because Anonymous FTP will allow just Read-Only access). The authorised network should be 192.168.0.0/16. Hit **Add** and then apply changes.

Now, we are fully done with that.

Section C: Joining the two together

Shift to the FTP server, open up a terminal and type in the following:

```
rm -rf /var/ftp/*
mount 192.168.1.250:/mnt/media /var/ftp -v
```

Create a test file in **/var/ftp**, connect to your FTP site with Firefox and see your handiwork in action.

That's all, folks.



Tip: To make that NFS share mount at system start-up, add the following line to **/etc/fstab**:

```
192.168.1.250:/mnt/media /var/ftp defaults 0 0
```

Replace **/mnt/media** with your own share's name. 

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